

Project Report: Global Seismic Risk Profiler

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1. Executive Summary

The Global Seismic Risk Profiler is an end-to-end data pipeline built on **Microsoft Fabric** that transforms raw earthquake telemetry from the **USGS API** into actionable business insights. By applying to the **Medallion Architecture (Bronze → Silver → Gold)**, the project delivers risk-ready datasets that support **insurance pricing, urban planning, and disaster preparedness**.

2. Business Problem

Raw seismic telemetry data is highly granular, inconsistent, and unsuitable for direct business use.

- **Granularity:** Individual earthquake events don't reveal long-term seismic trends.
- **Data Quality:** Raw feeds contain duplicates, nulls, and schema mismatches.
- **Business Readiness:** Analysts require datasets that translate technical signals into **risk categories, geospatial bins, and severity scores**.

3. Objectives

- Build a scalable pipeline to ingest and process USGS earthquake data.
- Standardize and clean data for consistency and reliability.
- Apply business logic to generate **risk-ready datasets**.
- Deliver insights through **SQL queries and Power BI dashboards**.

4. Methodology

4.1 Data Pipeline (Medallion Architecture)

- **Bronze Layer:** Ingest raw USGS JSON feeds for traceability.

- **Silver Layer:** Clean, transform, and consolidate data (remove duplicates, enforce schema).
- **Gold Layer:** Apply business rules (severity classification, geospatial aggregation, risk scoring).

4.2 Tools & Technologies

- **Microsoft Fabric** (Lakehouse, Data Engineering, Power BI)
- **PySpark** for ingestion and transformation
- **SQL** for analytical queries
- **Power BI** for visualization

5. Results

- **Cleaned Datasets:** Reliable, standardized seismic data.
- **Risk-Ready Tables:** Severity categories, geospatial bins, and risk scores.
- **Dashboards:** Interactive Power BI visuals showing seismic trends and risk zones.

6. Business Impact

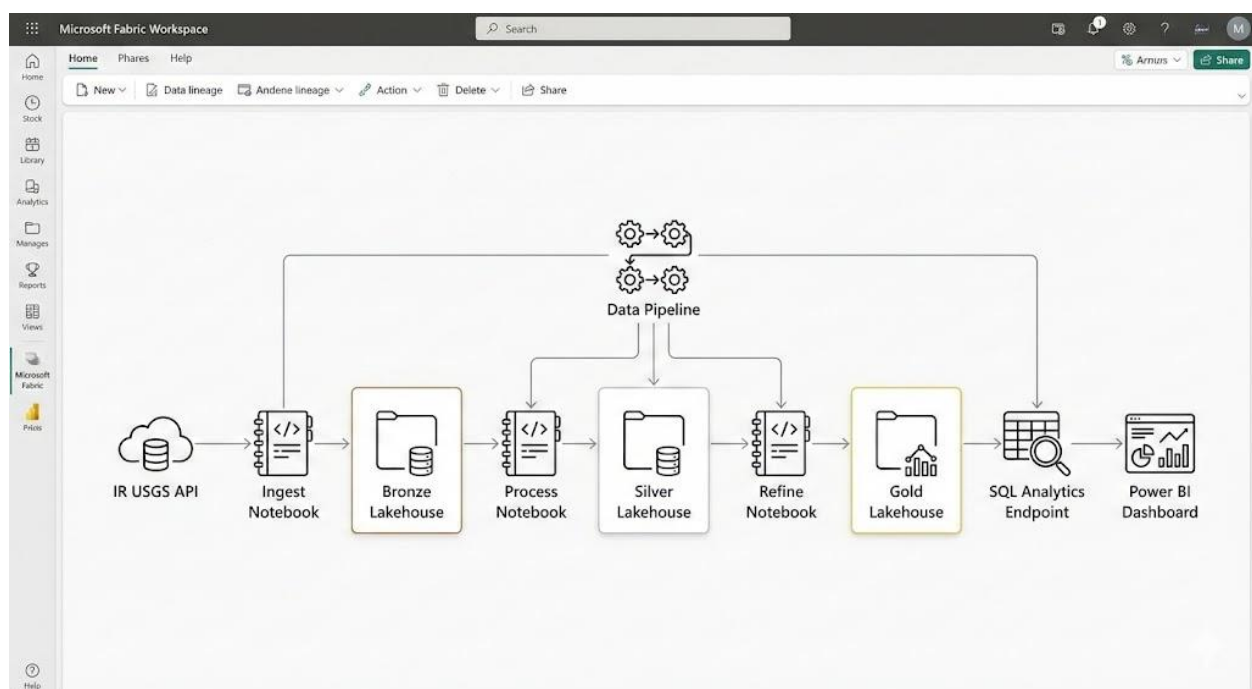
- **Insurance:** Enables accurate premium pricing in earthquake-prone regions.
- **Urban Planning:** Identifies high-risk zones for infrastructure development.
- **Disaster Preparedness:** Supports government agencies in resource allocation and response strategies.

7. Visuals

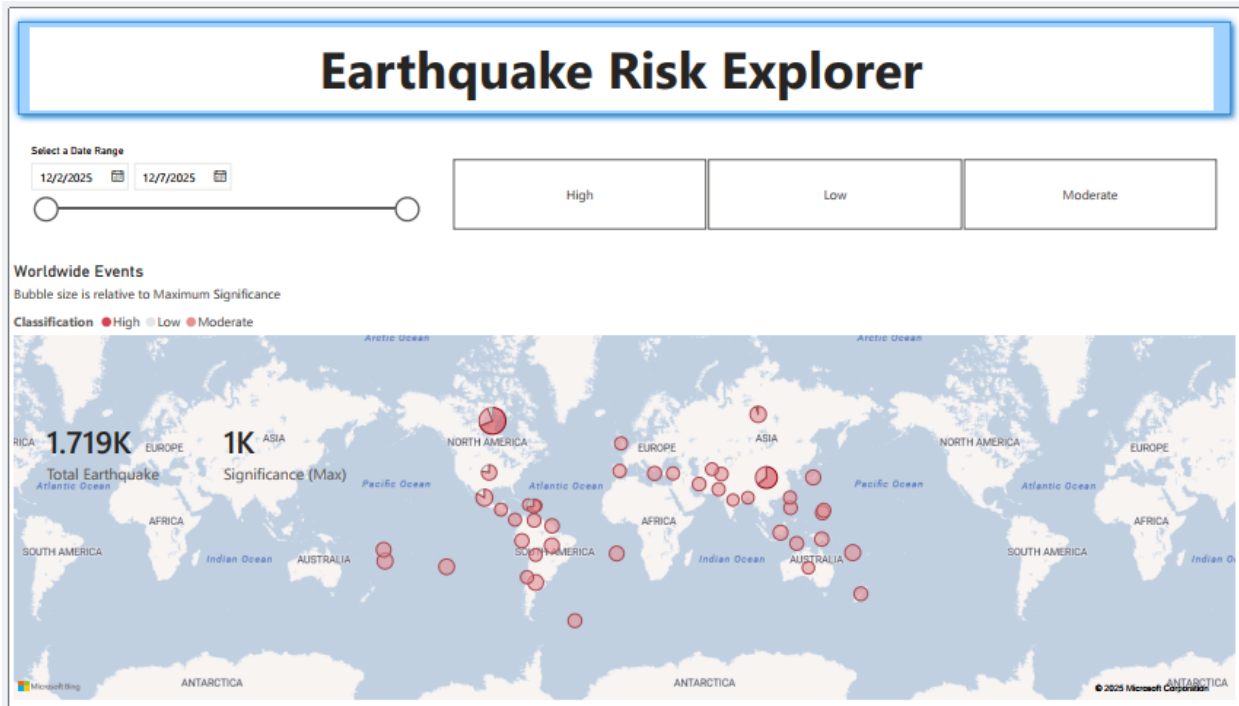
1. Architecture diagram



2. Lineage view



3. Power BI Dashboard :



8. Conclusion

The Global Seismic Risk Profiler demonstrates how raw telemetry can be transformed into **business-ready insights** using modern data engineering practices. This project showcases technical expertise in **Microsoft Fabric, PySpark, SQL, and Power BI**, while addressing a real-world business challenge with measurable impact.